

January 2014 Q7 C34

7.

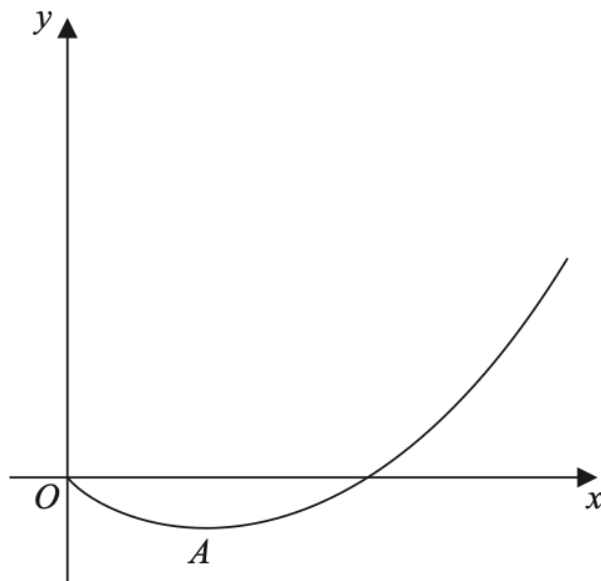
**Figure 3**

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = 2x(1+x) \ln x, \quad x > 0$$

The curve has a minimum turning point at A .

(a) Find $f'(x)$

(3)

(b) Hence show that the x coordinate of A is the solution of the equation

$$x = e^{-\frac{1+x}{1+2x}}$$

(3)

(c) Use the iteration formula

$$x_{n+1} = e^{-\frac{1+x_n}{1+2x_n}}, \quad x_0 = 0.46$$

to find the values of x_1 , x_2 and x_3 to 4 decimal places.

(3)

(d) Use your answer to part (c) to estimate the coordinates of A to 2 decimal places.

(2)

June 2014 Q1 C34

1. $f(x) = 2x^3 + x - 10$
- (a) Show that the equation $f(x) = 0$ has a root α in the interval $[1.5, 2]$ (2)
- The only real root of $f(x) = 0$ is α
- The iterative formula
- $$x_{n+1} = \left(5 - \frac{1}{2}x_n\right)^{\frac{1}{3}}, \quad x_0 = 1.5$$
- can be used to find an approximate value for α
- (b) Calculate x_1 , x_2 and x_3 , giving your answers to 4 decimal places. (3)
- (c) By choosing a suitable interval, show that $\alpha = 1.6126$ correct to 4 decimal places. (2)

10.

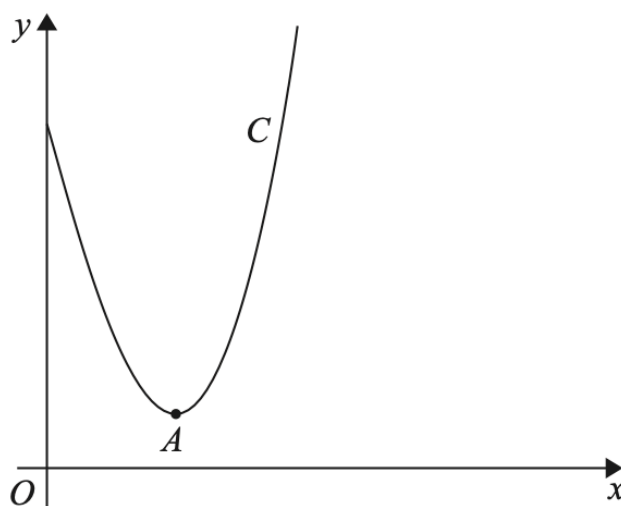


Figure 3

Figure 3 shows a sketch of part of the curve C with equation

$$y = \frac{x^2 \ln x}{3} - 2x + 4, \quad x > 0$$

Point A is the minimum turning point on the curve.

(a) Show, by using calculus, that the x coordinate of point A is a solution of

$$x = \frac{6}{1 + \ln(x^2)}$$

(5)

(b) Starting with $x_0 = 2.27$, use the iteration

$$x_{n+1} = \frac{6}{1 + \ln(x_n^2)}$$

to calculate the values of x_1 , x_2 and x_3 , giving your answers to 3 decimal places.

(3)

(c) Use your answer to part (b) to deduce the coordinates of point A to one decimal place.

(2)

January 2016 Q5 C34

5.

$$f(x) = -x^3 + 4x^2 - 6$$

- (a) Show that the equation $f(x) = 0$ has a root between $x = 1$ and $x = 2$

(2)

- (b) Show that the equation $f(x) = 0$ can be rewritten as

$$x = \sqrt{\left(\frac{6}{4-x}\right)}$$

(2)

- (c) Starting with $x_1 = 1.5$ use the iteration $x_{n+1} = \sqrt{\left(\frac{6}{4-x_n}\right)}$ to calculate the values of x_2 , x_3 and x_4 giving all your answers to 4 decimal places.

(3)

- (d) Using a suitable interval, show that 1.572 is a root of $f(x) = 0$ correct to 3 decimal places.

(2)

June 2016 Q10 C34

10. (a) Given that $-\frac{\pi}{2} < g(x) < \frac{\pi}{2}$, sketch the graph of $y = g(x)$ where

$$g(x) = \arctan x, \quad x \in \mathbb{R}$$

(2)

(b) Find the exact value of x for which

$$3g(x+1) - \pi = 0$$

(3)

The equation $\arctan x - 4 + \frac{1}{2}x = 0$ has a positive root at $x = \alpha$ radians.

(c) Show that $5 < \alpha < 6$

(2)

The iteration formula

$$x_{n+1} = 8 - 2 \arctan x_n$$

can be used to find an approximation for α

(d) Taking $x_0 = 5$, use this formula to find x_1 and x_2 , giving each answer to 3 decimal places.

(2)

January 2017 Q2 C34

2.

$$f(x) = x^3 - 5x + 16$$

- (a) Show that the equation $f(x) = 0$ can be rewritten as

$$x = (ax + b)^{\frac{1}{3}}$$

giving the values of the constants a and b .

(2)

The equation $f(x) = 0$ has exactly one real root α , where $\alpha = -3$ to one significant figure.

- (b) Starting with $x_1 = -3$, use the iteration

$$x_{n+1} = (ax_n + b)^{\frac{1}{3}}$$

with the values of a and b found in part (a), to calculate the values of x_2 , x_3 and x_4 , giving all your answers to 3 decimal places.

(3)

- (c) Using a suitable interval, show that $\alpha = -3.17$ correct to 2 decimal places.

(2)

October 2017 Q1 C34

1. $f(x) = x^5 + x^3 - 12x^2 - 8, \quad x \in \mathbb{R}$
- (a) Show that the equation $f(x) = 0$ can be written as
- $$x = \sqrt[3]{\frac{4(3x^2 + 2)}{x^2 + 1}} \quad (3)$$
- (b) Use the iterative formula
- $$x_{n+1} = \sqrt[3]{\frac{4(3x_n^2 + 2)}{x_n^2 + 1}}$$
- with $x_0 = 2$, to find x_1, x_2 and x_3 giving your answers to 3 decimal places. (3)
- The equation $f(x) = 0$ has a single root, α .
- (c) By choosing a suitable interval, prove that $\alpha = 2.247$ to 3 decimal places. (2)

January 2018 Q3 C34

3.
$$f(x) = \frac{x^2}{4} + \ln(2x), \quad x > 0$$

(a) Show that the equation $f(x) = 0$ can be rewritten as

$$x = \frac{1}{2}e^{-\frac{1}{4}x^2}$$

(2)

The equation $f(x) = 0$ has a root near 0.5

(b) Starting with $x_1 = 0.5$ use the iterative formula

$$x_{n+1} = \frac{1}{2}e^{-\frac{1}{4}x_n^2}$$

to calculate the values of x_2 , x_3 and x_4 , giving your answers to 4 decimal places.

(3)

(c) Using a suitable interval, show that 0.473 is a root of $f(x) = 0$ correct to 3 decimal places.

(2)

June 2018 Q3 C34

3. $f(x) = 2^{x-1} - 4 + 1.5x \quad x \in \mathbb{R}$

(a) Show that the equation $f(x) = 0$ can be written as

$$x = \frac{1}{3}(8 - 2^x) \quad (2)$$

The equation $f(x) = 0$ has a root α , where $\alpha = 1.6$ to one decimal place.

(b) Starting with $x_0 = 1.6$, use the iteration formula

$$x_{n+1} = \frac{1}{3}(8 - 2^{x_n})$$

to calculate the values of x_1 , x_2 and x_3 , giving your answers to 3 decimal places.

(3)

(c) By choosing a suitable interval, prove that $\alpha = 1.633$ to 3 decimal places.

(2)

4.

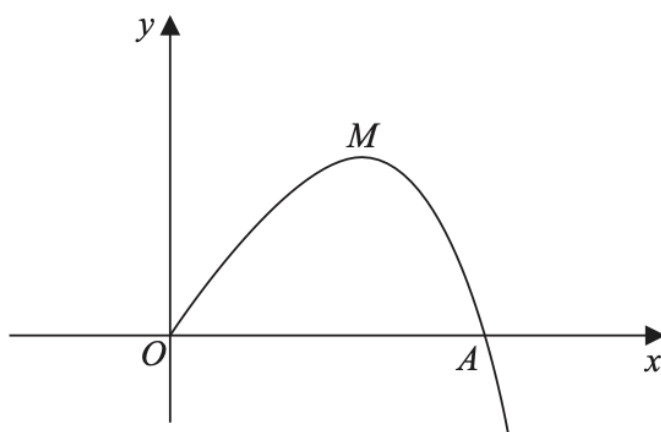


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = 8x - xe^{3x}$, $x \geq 0$

The curve meets the x -axis at the origin and cuts the x -axis at the point A .

(a) Find the exact x coordinate of A , giving your answer in its simplest form.

(2)

The curve has a maximum turning point at the point M .

(b) Show, by using calculus, that the x coordinate of M is a solution of

$$x = \frac{1}{3} \ln \left(\frac{8}{1 + 3x} \right)$$

(5)

(c) Use the iterative formula

$$x_{n+1} = \frac{1}{3} \ln \left(\frac{8}{1 + 3x_n} \right)$$

with $x_0 = 0.4$ to calculate the values of x_1 , x_2 and x_3 , giving your answers to 3 decimal places.

(3)

January 2019 Q11 C34

11. (a) Given that $0 \leq f(x) \leq \pi$, sketch the graph of $y = f(x)$ where

$$f(x) = \arccos(x - 1), \quad 0 \leq x \leq 2 \quad (2)$$

The equation $\arccos(x - 1) - \tan x = 0$ has a single root α .

- (b) Show that $0.9 < \alpha < 1.1$ (2)

The iteration formula

$$x_{n+1} = \arctan(\arccos(x_n - 1))$$

can be used to find an approximation for α .

- (c) Taking $x_0 = 1.1$ find, to 3 decimal places, the values of x_1 and x_2 (2)

June 2019 Q1 C34

1.

$$f(x) = 2x^3 + x - 20$$

- (a) Show that the equation $f(x) = 0$ can be rewritten as

$$x = \sqrt[3]{a - bx}$$

where a and b are positive constants to be determined.

(2)

- (b) Starting with $x_1 = 2.1$ use the iteration formula $x_{n+1} = \sqrt[3]{a - bx_n}$, with the numerical values of a and b , to calculate the values of x_2 and x_3 giving your answers to 3 decimal places.

(2)

- (c) Using a suitable interval, show that 2.077 is a root of the equation $f(x) = 0$ correct to 3 decimal places.

(2)

- (d) Hence state a root, to 3 decimal places, of the equation

$$2(x + 2)^3 + x - 18 = 0$$

(1)

7.

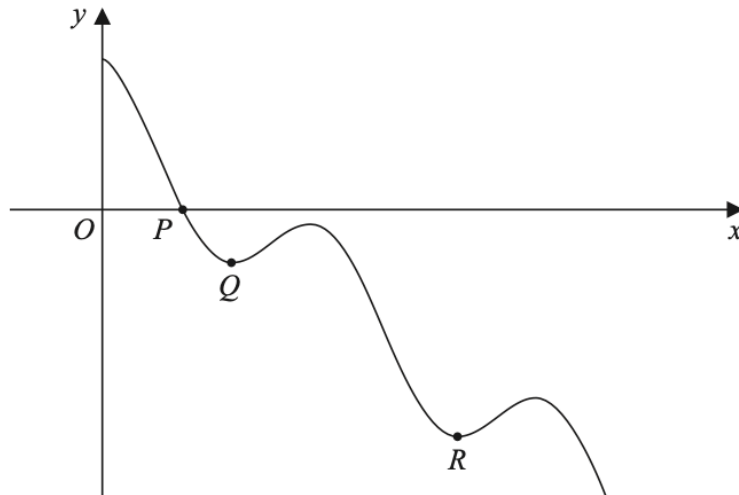


Figure 3

Figure 3 shows a sketch of part of the curve with equation

$$y = 2 \cos 3x - 3x + 4 \quad x > 0$$

where x is measured in radians.

The curve crosses the x -axis at the point P , as shown in Figure 3.

Given that the x coordinate of P is α ,

(a) show that α lies between 0.8 and 0.9

(2)

The iteration formula

$$x_{n+1} = \frac{1}{3} \arccos(1.5x_n - 2)$$

can be used to find an approximate value for α .

(b) Using this iteration formula with $x_1 = 0.8$ find, to 4 decimal places, the value of

(i) x_2

(ii) x_5

(3)

The point Q and the point R are local minimum points on the curve, as shown in Figure 3.

Given that the x coordinates of Q and R are β and λ respectively, and that they are the two smallest values of x at which local minima occur,

(c) find, using calculus, the exact value of β and the exact value of λ .

(6)

6.

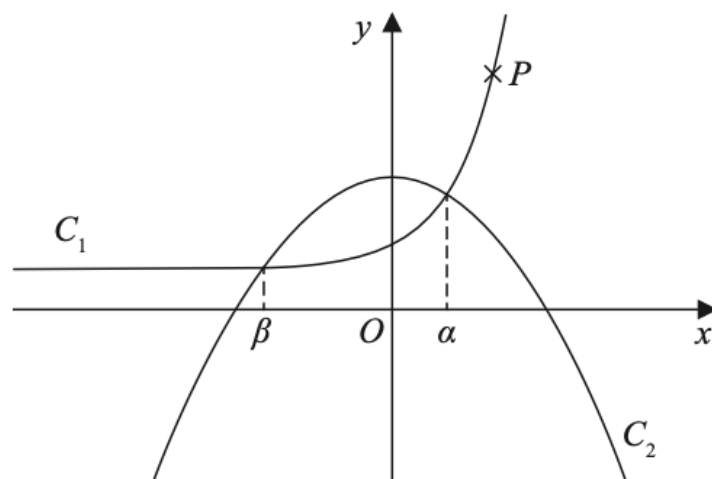


Figure 3

Figure 3 shows a sketch of curve C_1 with equation $y = 5e^{x-1} + 3$

and curve C_2 with equation $y = 10 - x^2$

The point P lies on C_1 and has y coordinate 18

- (a) Find the x coordinate of P , writing your answer in the form $\ln k$, where k is a constant to be found.

(3)

The curve C_1 meets the curve C_2 at $x = \alpha$ and at $x = \beta$, as shown in Figure 3.

- (b) Using a suitable interval and a suitable function that should be stated, show that to 3 decimal places $\alpha = 1.134$

(3)

The iterative equation

$$x_{n+1} = -\sqrt{7 - 5e^{x_n-1}}$$

is used to find an approximation to β .

Using this iterative formula with $x_1 = -3$

- (c) find the value of x_2 and the value of β , giving each answer to 6 decimal places.

(3)

January 2021 Q6 P3

6.

$$f(x) = x \cos\left(\frac{x}{3}\right) \quad x > 0$$

(a) Find $f'(x)$

(2)

(b) Show that the equation $f'(x) = 0$ can be written as

$$x = k \arctan\left(\frac{k}{x}\right)$$

where k is an integer to be found.

(2)

(c) Starting with $x_1 = 2.5$ use the iteration formula

$$x_{n+1} = k \arctan\left(\frac{k}{x_n}\right)$$

with the value of k found in part (b), to calculate the values of x_2 and x_6 giving your answers to 3 decimal places.

(2)

(d) Using a suitable interval and a suitable function that should be stated, show that a root of $f'(x) = 0$ is 2.581 correct to 3 decimal places.

(2)

9.

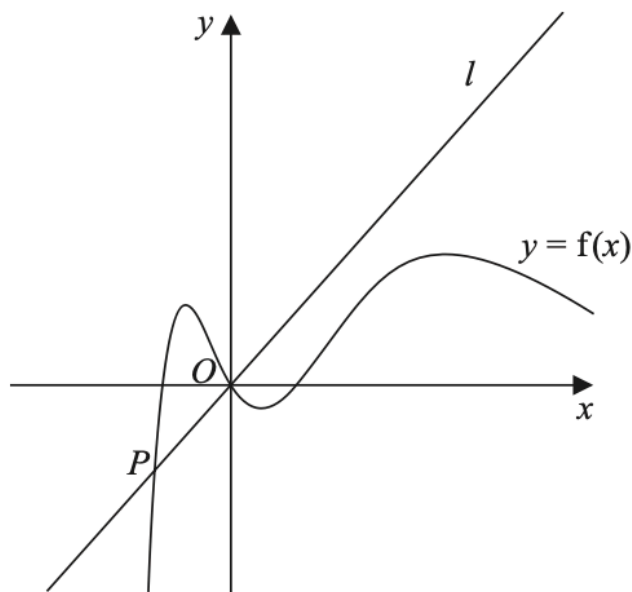


Figure 3

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = x(x^2 - 4)e^{-\frac{1}{2}x}$$

(a) Find $f'(x)$.

(2)

The line l is the normal to the curve at O and meets the curve again at the point P .

The point P lies in the 3rd quadrant, as shown in Figure 3.

(b) Show that the x coordinate of P is a solution of the equation

$$x = -\frac{1}{2}\sqrt{16 + e^{\frac{1}{2}x}}$$

(4)

(c) Using the iterative formula

$$x_{n+1} = -\frac{1}{2}\sqrt{16 + e^{\frac{1}{2}x_n}} \quad \text{with } x_1 = -2$$

find, to 4 decimal places,

(i) the value of x_2

(ii) the x coordinate of P .

(3)

9.

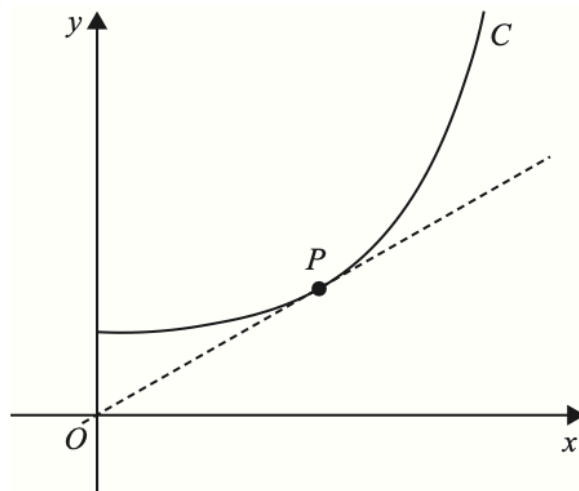


Figure 3

Figure 3 shows a sketch of part of the curve C with equation

$$y = \sqrt{3 + 4e^{x^2}} \quad x \geq 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form. (2)

The point P with x coordinate α lies on C .

Given that the tangent to C at P passes through the origin, as shown in Figure 3,

- (b) show that $x = \alpha$ is a solution of the equation (3)
- $$4x^2e^{x^2} - 4e^{x^2} - 3 = 0$$

- (c) Hence show that α lies between 1 and 2 (2)

- (d) Show that the equation in part (b) can be written in the form

$$x = \frac{1}{2}\sqrt{4 + 3e^{-x^2}} \quad (1)$$

The iteration formula

$$x_{n+1} = \frac{1}{2}\sqrt{4 + 3e^{-x_n^2}}$$

with $x_1 = 1$ is used to find an approximation for α .

- (e) Use the iteration formula to find, to 4 decimal places, the value of

(i) x_3

(ii) α

(3)

June 2023 Q1 P3

1. $g(x) = x^6 + 2x - 1000$

- (a) Show that $g(x) = 0$ has a root α in the interval $[3, 4]$

(2)

Using the iteration formula

$$x_{n+1} = \sqrt[6]{1000 - 2x_n} \quad \text{with } x_1 = 3$$

- (b) (i) find, to 4 decimal places, the value of x_2

- (ii) find, by repeated iteration, the value of α .
Give your answer to 4 decimal places.

(3)

October 2023 Q1 P3

1. A curve has equation $y = f(x)$ where

$$f(x) = x^2 - 5x + e^x \quad x \in \mathbb{R}$$

(a) Show that the equation $f(x) = 0$ has a root, α , in the interval $[1, 2]$

(2)

The iterative formula

$$x_{n+1} = \sqrt{5x_n - e^{x_n}}$$

with $x_1 = 1$ is used to find an approximate value for the root α .

(b) (i) Find the value of x_2 to 4 decimal places.

(ii) Find, by repeated iteration, the value of α , giving your answer to 4 decimal places.

(3)

January 2024 Q2 P3

2. A curve has equation $y = f(x)$ where

$$f(x) = x^4 - 5x^2 + 4x - 7 \quad x \in \mathbb{R}$$

(a) Show that the equation $f(x) = 0$ has a root, α , in the interval $[2, 3]$

(2)

(b) Show that the equation $f(x) = 0$ can be written as

$$x = \sqrt[3]{\frac{5x^2 - 4x + 7}{x}}$$

(1)

The iterative formula

$$x_{n+1} = \sqrt[3]{\frac{5x_n^2 - 4x_n + 7}{x_n}}$$

is used to find α

(c) Starting with $x_1 = 2$ and using the iterative formula,

(i) find, to 4 decimal places, the value of x_2

(ii) find, to 4 decimal places, the value of α

(3)